LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to Engine indicating and crew alerting system (EICAS) & Electronic centralized Aircraft Monitoring (ECAM)
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to Engine indicating and crew alerting system (EICAS) & Electronic centralized Aircraft Monitoring (ECAM). Practical on EICAS & ECAM.
5. Specific Objectives	20.7.1.1 Trainee Will Be able to explain electronic displays of engine performance parameters.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on EICAS & ECAM: Demonstrate engine parameter display selection, verify alert message hierarchy, and simulate normal/abnormal mode transitions under instructor supervision.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the EICAS system?
(b) How does ECAM differ from traditional analog engine instruments?
(c) List three common engine performance parameters displayed electronically.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• EICAS System Architecture Overview • Engine Performance Parameter Displays • Alert Message Hierarchy and Priority • Display Modes: Normal and Backup • Crew Alerting System Integration	L & Ds	PPT, IB	15 min
(b)	• ECAM System Architecture Overview • Normal and Abnormal Display Modes • Engine Parameter Monitoring on ECAM • Alert Message Categories and Actions • Comparison between EICAS and ECAM	L & Ds	PPT, PDF	20 min
(c)	• Practical Demonstration of EICAS Display • Practical Demonstration of ECAM Display • Simulation of Alert Message Scenarios • Verification of Crew Alerting Functions • Instructor Supervision and Feedback	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference in alert message priority between EICAS and ECAM systems.
(b) Describe how engine performance parameters are presented in normal versus backup modes.
(c) Identify the primary advantage of electronic centralized monitoring over analog systems.

12. Home Assignment and Follow Up	Prepare a brief note comparing the display units and alert message handling of EICAS and ECAM for the next lesson (LP 18).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	EICAS Display units, Display modes, Display select panels and alert messages.	Refer to the master syllabus reference list	LP 18

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min